fn make_noisy_top_k

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October 11, 2026

Proves soundness of make_noisy_top_k in mod.rs at commit f5bb719 (outdated[1]).
make_noisy_top_k returns a Measurement that noisily selects the indices of the greatest scores from a vector of input scores.

1 Hoare Triple

Precondition

Compiler-verified

- MO is a type with trait TopKMeasure
- TIA (atomic input type) is a type with trait Number

Caller-verified

None

Pseudocode

```
def make_noisy_top_k(
    input_domain: VectorDomain[AtomDomain[TIA]],
    input_metric: LInfDistance[TIA],
    privacy_measure: MO,
    k: usize,
    scale: f64,
    negate: bool,
) -> Measurement:
    if input_domain.element_domain.nan():  #
        raise "input domain elements must be non-nan"

    if input_domain.size is not None:
        if k > input_domain.size:
            raise "k must not exceed the number of candidates"

    if (
        not scale.is_finite() or scale.is_sign_negative()
    ):  #
        raise "scale must be finite and non-negative"

    monotonic = input_metric.monotonic

    def privacy_map(d_in: TIA):  #
        # convert to range distance
```

[1]See new changes with git diff f5bb719..12d5646 rust/src/measurements/noisy_top_k/mod.rs

```
25          d_in = d_in if monotonic else d_in.inf_add(d_in)
26          d_in = f64.inf_cast(d_in)  #
27
28          if d_in.is_sign_negative():  #
29              raise "sensitivity must be non-negative"
30
31          if d_in.is_zero():  #
32              return 0.0
33
34          if scale.is_zero():  #
35              return f64.INFINITY
36
37          #
38          return MO.privacy_map(d_in, scale).inf_mul(f64.inf_cast(k))
39
40      return Measurement.new(
41          input_domain=input_domain,
42          input_metric=input_metric,
43          output_measure=privacy_measure,
44          function=lambda x: noisy_top_k(x, scale, k, negate, MO.REPLACEMENT),
45          privacy_map=privacy_map,
46      )
```

Postcondition

Theorem 1.1. For every setting of the input parameters `input_domain, input_metric, output_measure, k, scale, negate, MO, TIA` to `make_noisy_top_k` such that the given preconditions hold, `make_noisy_top_k` raises an error (at compile time or run time) or returns a valid measurement. A valid measurement has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke`(x) and `invoke`(x') either both return the same error or neither return an error.

2. (Privacy guarantee). For every pair of members x and x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for
   `output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map`(`d_in`) does not raise an error, and `privacy_map`(`d_in`) $=$ `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_measure`.

Proof of data-independent errors. By the postcondition of `noisy_top_k`, the only source of error is due to entropy exhaustion, which could be data-dependent, due to differing number of expected random draws depending on the input dataset.

Therefore, the mechanism only satisfies the requirement for data-independent errors when conditioned on entropy not being exhausted. □

Proof of privacy guarantee. When `d_in` is zero, by line 31, the privacy loss is zero, satisfying the postcondition. Otherwise when scale is zero, by line 34, the privacy loss is infinite, also satisfying the postcondition.

By the checks on lines 9 and 18, the preconditions for `noisy_top_k` are satisfied. Additionally by the checks on lines 28, 31 and 34, the preconditions for `TopKMeasure``privacy_map` are satisfied.

By the postcondition of `TopKMeasure` and adaptive composition, the `d_out`on line 37 satisfies the postcondition. □